ASSIGNMENT REPORT

Oriented Object Programming - SCC0504

Davi Possato Virtudes - 16861780
Aron Costa Araújo - 16855150
Richard William

1. Requirements

We covered all the fundamental requirements and didn’t add any extra features.

2. Project Description

We made a desktop calendar and event management application where users can create, edit, and delete events such as meetings, birthdays, and appointments; view them in a monthly calendar; and receive in-app reminders before events start.

We applied Java Swing to build a multi-component GUI, implement event-driven programming, manage structured data with file persistence, and handle scheduling logic, including recurring events.

3. Test Plan

To guarantee the system's stability and resilience, the testing protocol prioritized comprehensive input validation. For every user entry point, testers systematically introduced a wide array of data types, covering:

- **Standard Inputs:** Valid data types (e.g., integers, strings, floats) expected during typical application usage.
- **Boundary Conditions:** Extreme or maximum/minimum values to evaluate how the system handles thresholds.
- **Invalid and Unexpected Data:** Mismatched data types, special characters, malicious injections, and null/empty fields to ensure the application fails gracefully without crashing.

We tested step-by-step the implementation of every feature. And tested the final version, to make sure that everything was working as expected together.

4. Test Result

The execution of our quality assurance strategy yielded highly successful results across all deployment metrics. To evaluate these outcomes accurately, a clear distinction must be made between active debugging and formal validation. Throughout the development lifecycle, test failures were encountered exclusively during the initial debugging phases; these were intentional, controlled failures utilized by the team to isolate bugs and refine the codebase.

At every stage of implementation, as each feature reached functional completeness, the corresponding final validation tests passed with the expected output, guaranteeing the success of the implementation.

5. Build Procedures

To run our code, we strongly recommend using the IntelliJ IDE, but there's many ways to run it. We will assume that the user will use IntelliJ and is using Windows.

To run the app using IntelliJ you must follow the following instructions:

1 - Install JDK:
To run, is essential to have the JDK (version 11 or higher). To install it, you must go to the official Java Downloading Page. There you must find the version you want to install and the link to download according to your operating system (Linux, Mac or Windows).
As you installed the archive from the web page, you must treat it. You can have installed the installer file or the compressed archive. If you installed the installer file, you must run it with the administrator privileges. If you download the compressed archive, you must unzip it in the java default directory of your computer (generally C:/Programs Archives/Java).

2 - Download the project:
You can find the project on this link. You must click on "code" and "download zip".
After downloading the zip, you must unzip the zip file on your computer.

3 - Setting the environment variables
Open the System Proprieties. Click on Advanced and Environment Variables. Find the "PATH" variable and double click on it. Click on "new" and paste the JDK path.

4- Install the IntelliJ IDE
You can download the IntelliJ IDE on this link. When you go to the Official IntelliJ Download Page,you must chose your operating system and the format of the downloading file.
If you chose to install the installer file, you must run it using administrator privileges.
After installing the IntelliJ on your computer, you must follow the instructions and configure it as you like.

5 - Opening and running the application
In the IntelliJ interface, you must see some "open" button. Click on it. Then you may select the project folder that you unzipped.
After that, you must find Main.java file
(C:\Users\davip\Downloads\1\EventPlanner-main\src\eventplanner) and select it by double clicking.
Finally, you just need to run it by clicking on the "play" button or pressing shift+f10.

6. Problems

Overall, the development of the application was highly stable, with the team navigating routine technical hurdles without encountering any extraordinary or disruptive anomalies.

The most notable challenges emerged within frontend engineering and user interface (UI) design. Adapting to the specialized frontend tooling and frameworks presented a learning curve, requiring dedicated effort to make sure that we were developing the best interface possible and covering all the requirements established.

7. Comments

The project was a great success. It was a very practical and nice way to reinforce learning and guarantee that the course content was absorbed by all of us.